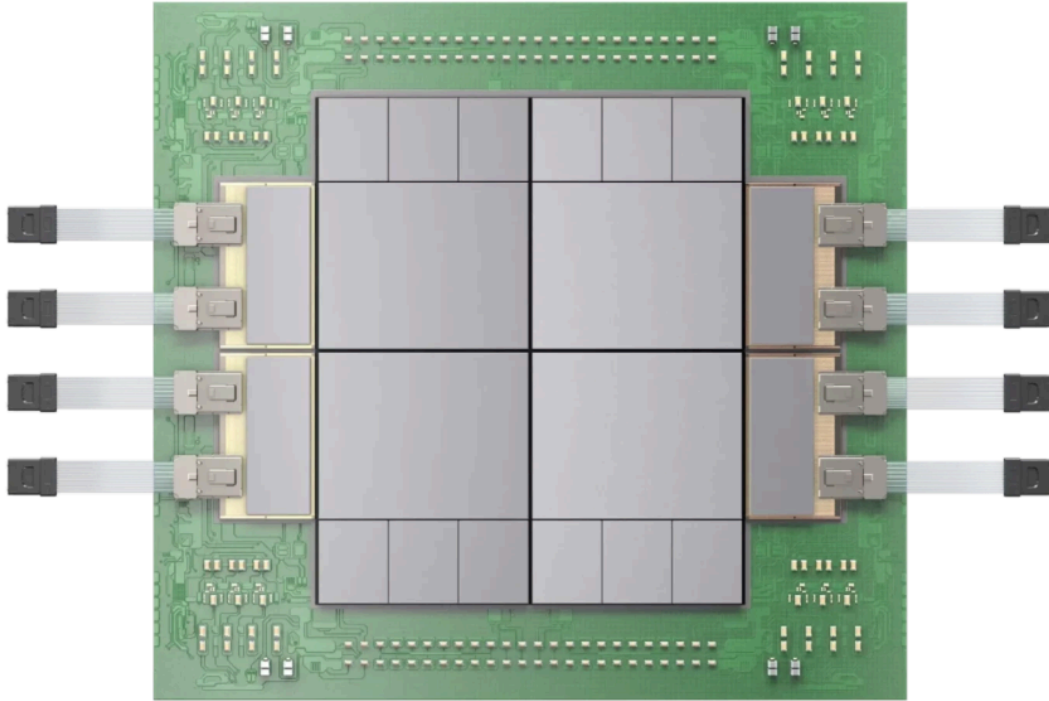




Passage L200 Specifications

Passage™ L200

The first Edgeless I/O™ 3D Co-Packaged Optics, eliminating the shoreline bottleneck to deliver 32–64 Tbps of optical I/O per chip.



Bandwidth	Total I/O	Density	Latency
32–64 Tbps Optical I/O	200+ Tbps / Chip	10x vs Pluggables	Linear Drive Architecture

BREAKING THE SHORELINE CONSTRAINT

The Challenge

Conventional chip I/O is confined to the die edges, known in the industry as the shoreline. As die area scales with the square of the radius, perimeter scales only linearly, creating a bandwidth bottleneck. Chips exhaust edge space long before they exhaust compute capability.

Passage L200 Implementation

Passage™ L200 removes this constraint by vertically integrating the photonic IC (PIC) beneath the electronic IC (EIC). This places SerDes and optical interfaces across the entire die surface, enabling bandwidth density that scales with silicon area rather than die perimeter.

Technical Specifications

Complete optical, integration, and performance specifications for Passage L200 and Passage L200X.

OPTICAL

• Bandwidth

- 32 Tbps optical I/O (Passage L200)
- 64 Tbps optical I/O (Passage L200X)
- 200+ Tbps total I/O per chip

• Modulation

- PAM4 at 56G and 112G

• Energy Efficiency

- <5 pJ/bit (optical)

• Reach

- 10m to 2km (DR/FR direct drive)

• Topology

- Edgeless I/O area-array

• Serviceability

- Detachable fiber connectors

INTEGRATION

• Architecture

- 3D vertical integration
- Chip-on-Wafer (CoW) bonding
- Photonic IC (PIC) stacked beneath electronic IC (EIC)

• I/O Topology

- Edgeless I/O — SerDes and optical interfaces distributed across entire die surface
- Bandwidth density scales with silicon area, not die perimeter

• Drive Architecture

- Linear drive — eliminates DSP latency

PERFORMANCE

• Bandwidth Density

- 5–10x greater than conventional pluggable solutions
- Scales with die area rather than shoreline perimeter

• Latency

- Linear drive architecture — no DSP re-timing overhead

• Power

- <5 pJ/bit optical energy efficiency

Product Variants

Passage L200 is available in two variants optimized for different bandwidth and infrastructure requirements.

	Passage L200	Passage L200X
Optical I/O	32 Tbps	64 Tbps
Target Workload	Next-generation XPU's requiring breakthrough bandwidth density	Ultra-dense AI infrastructure with maximum bandwidth demand
Integration	3D Vertical Integration, Chip-on-Wafer (CoW)	3D Vertical Integration, Chip-on-Wafer (CoW)
Fiber	Detachable	Detachable
Bandwidth Density	5–10x vs pluggables	2x vs Passage L200 (10–20x vs pluggables)
Scalability	Standard for 3D co-packaged optics	Extreme scalability for demanding workloads

AVAILABILITY

Passage L200 design partnerships are open for 2026 roadmaps. Contact Lightmatter sales to discuss integration requirements and qualification timelines.